

MSCR Tracker — Performance Analytics

Session: 1227 frames | 73.8 s | Resolution 848×480 @ 30 fps

2.4 %

3-D Detection
Rate

83.7 %

Skeleton
Found Rate

51.4 ms

Mean Proc.
Time

19.4 fps

Throughput
(est.)

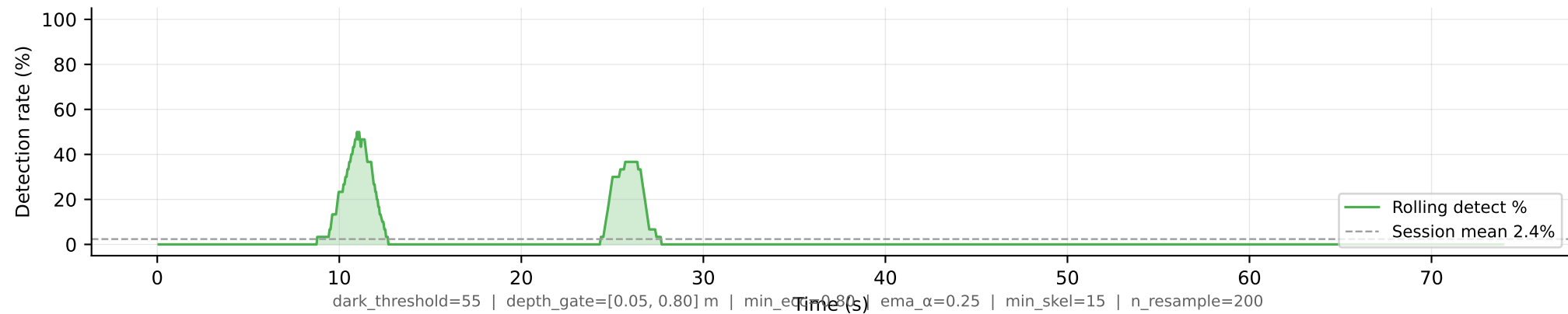
35.4 mm

Mean Arc
Length

5.15 mm

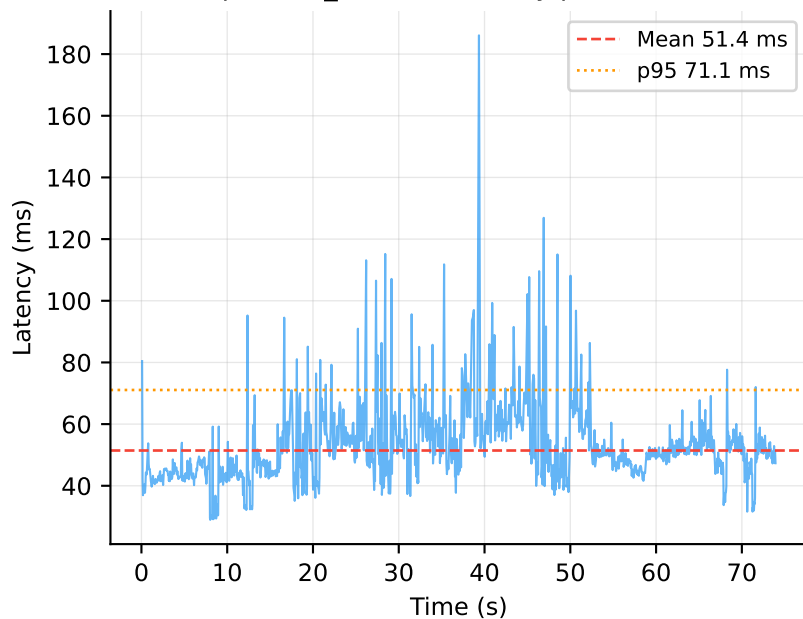
Tip Jitter
(p95)

Rolling 3-D Detection Rate (30-frame window)

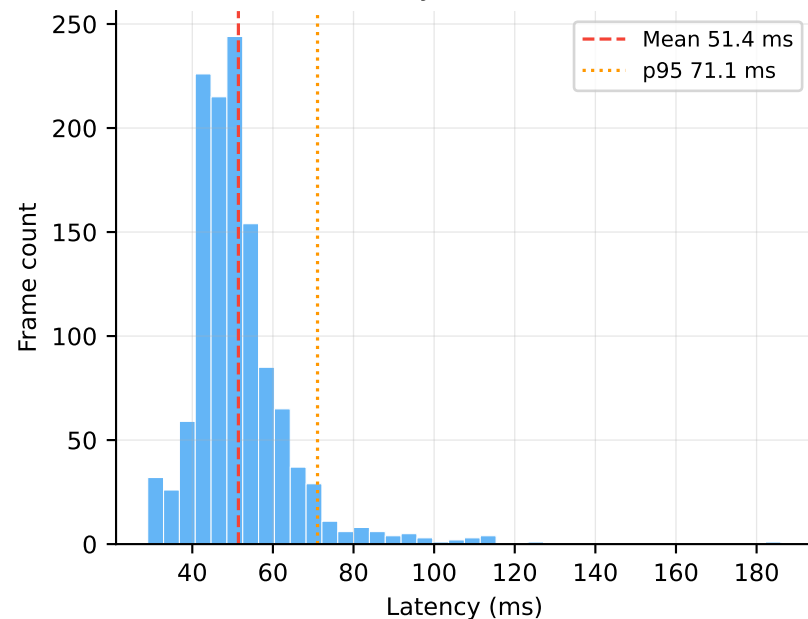


Processing Latency & Skeleton Quality

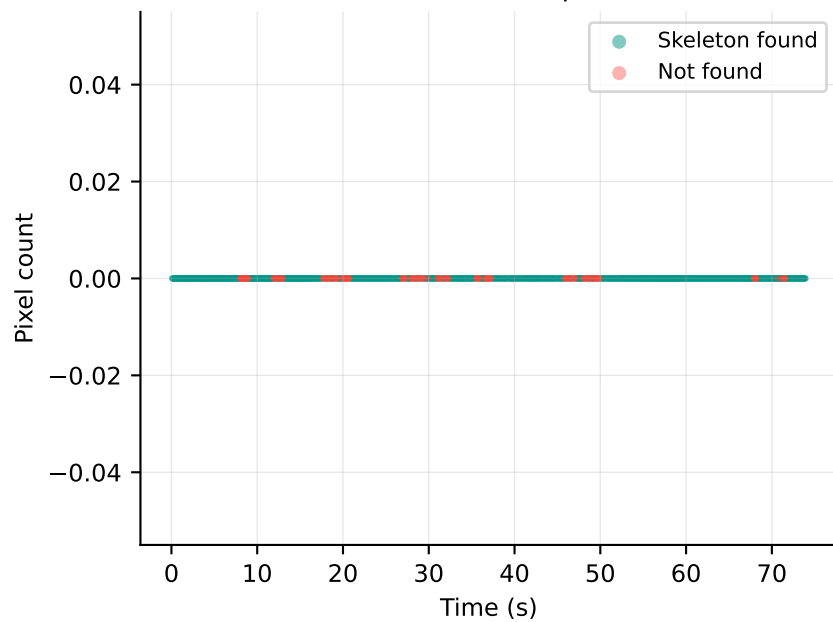
process_frame() Latency per Frame



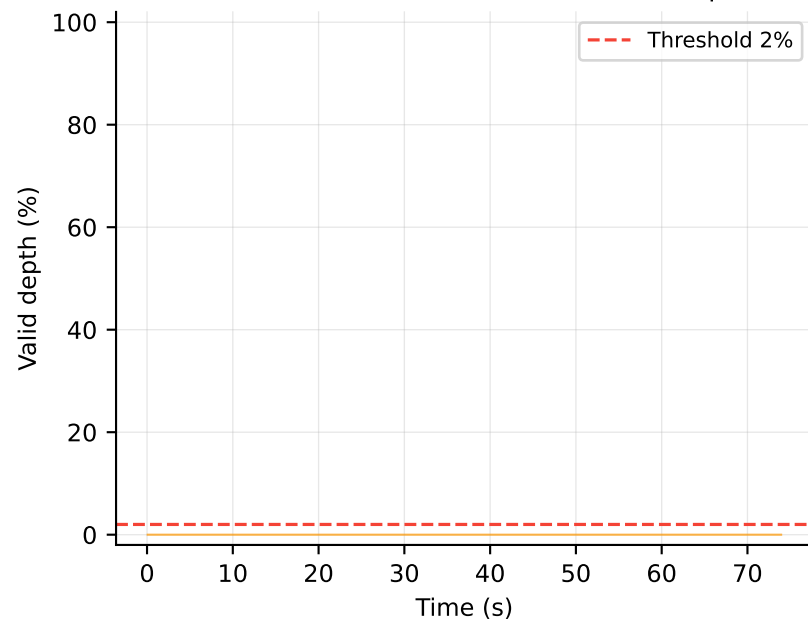
Latency Distribution



Skeleton Pixel Count per Frame

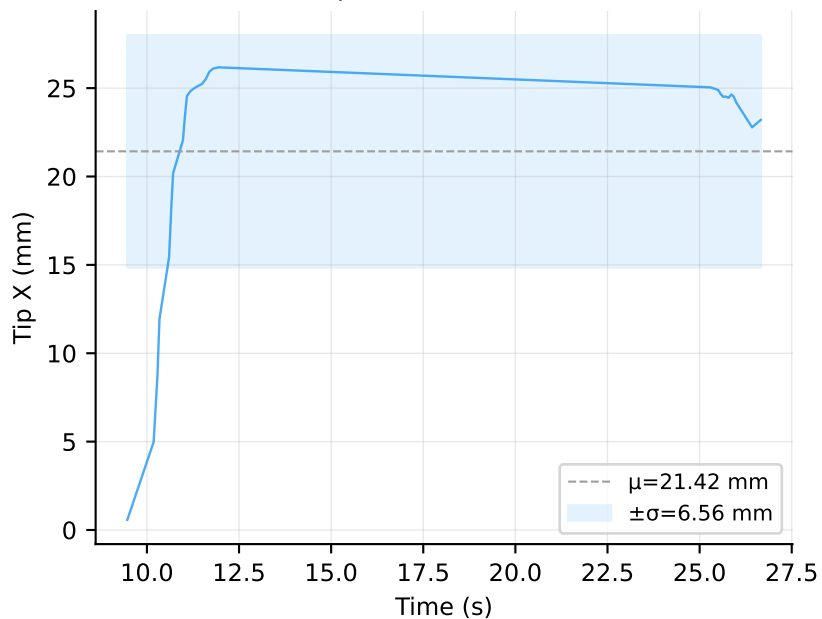


Fraction of Skeleton Pixels with Valid Depth

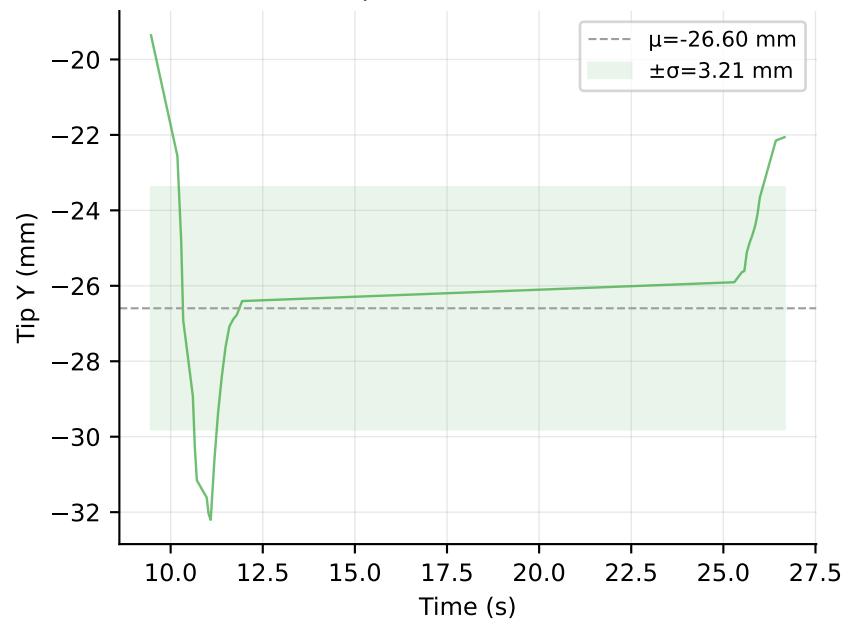


3-D Tip Position Trajectories

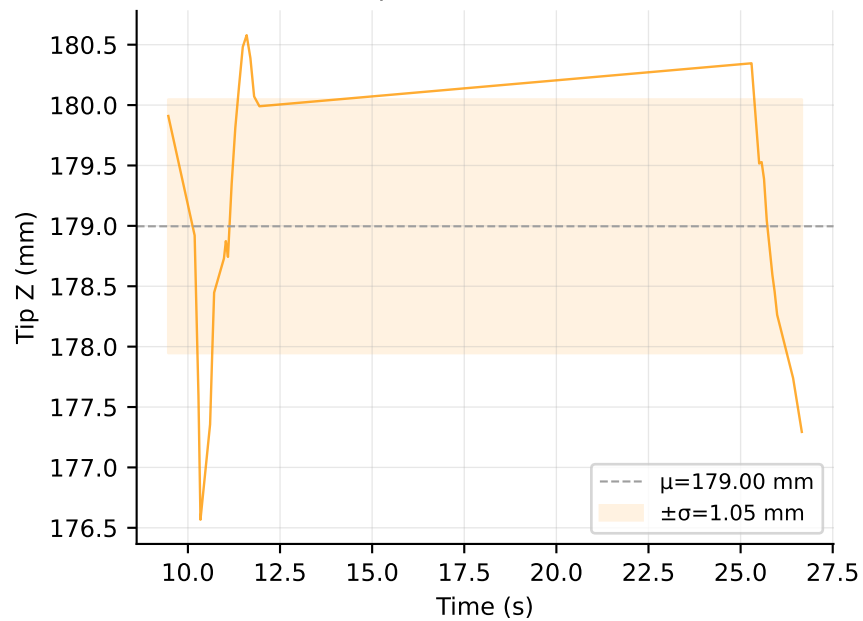
Tip X (mm) vs Time



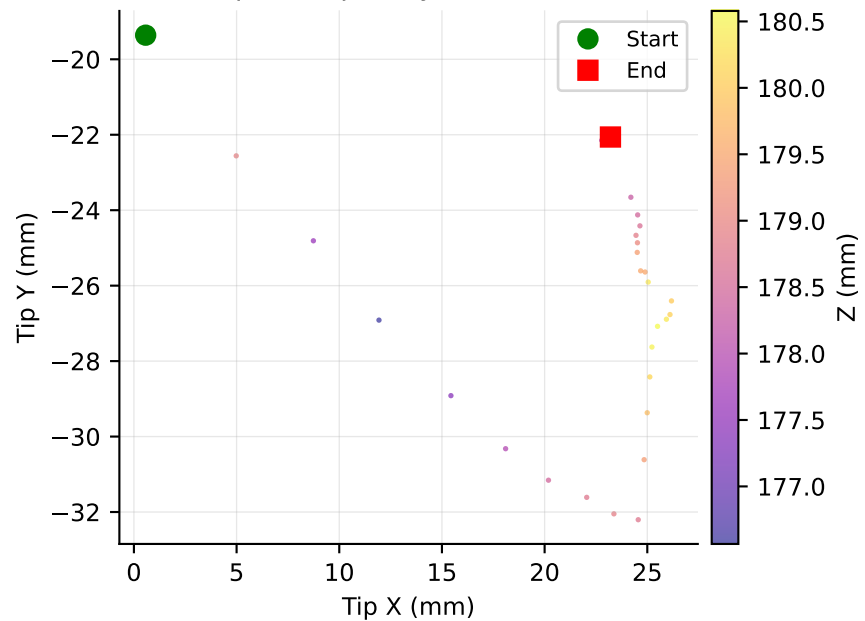
Tip Y (mm) vs Time



Tip Z (mm) vs Time

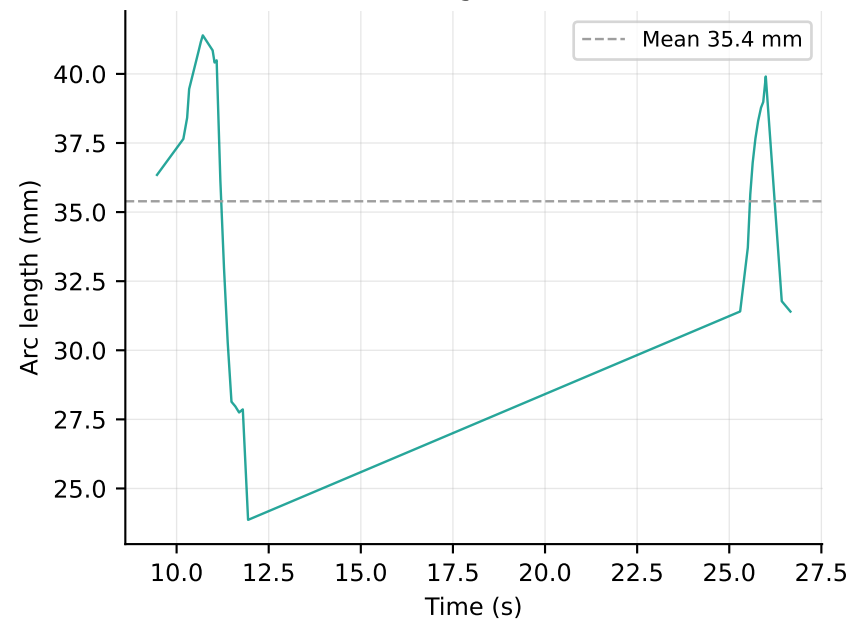


Tip XY Trajectory (colour = Z)

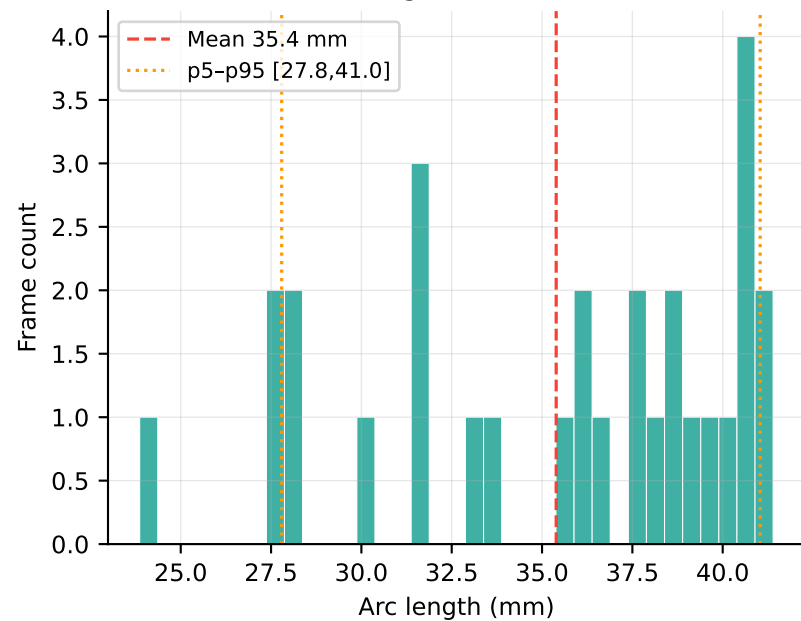


Arc Length & Tip Stability

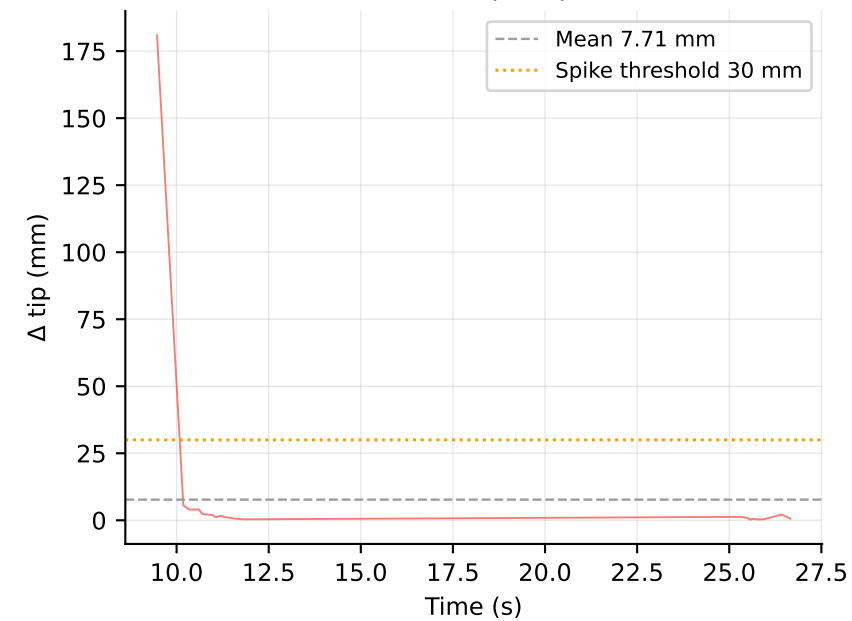
3-D Arc Length over Time



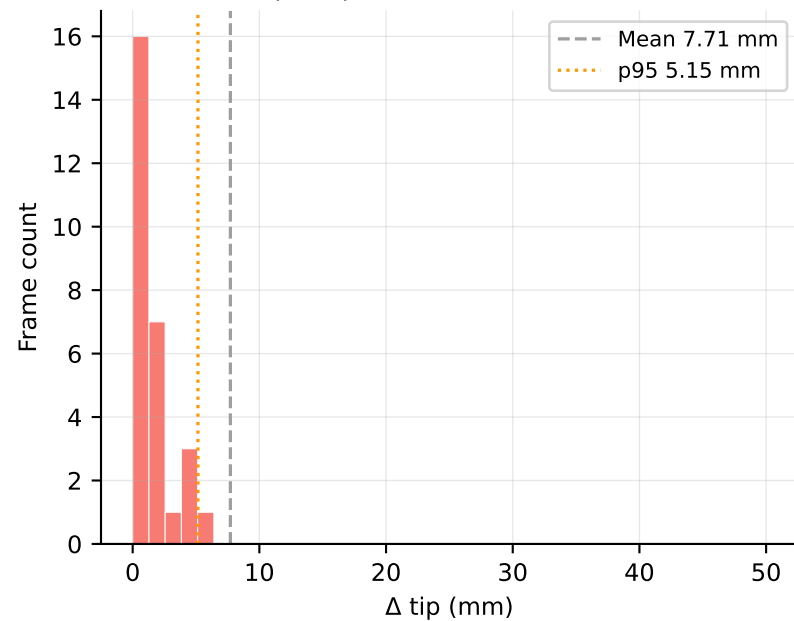
Arc Length Distribution



Frame-to-Frame Tip Displacement



Tip Displacement Distribution



Summary Statistics Table

Metric	Mean	Std	Min	Max	p5	p95	Unit
Process latency	51.437	12.373	28.942	186.085	37.012	71.070	ms
Skeleton pixel count	0.000	0.000	0.000	0.000	0.000	0.000	px
Valid depth fraction	0.000	0.000	0.000	0.000	0.000	0.000	%
Arc length	35.391	4.941	23.864	41.395	27.794	41.037	mm
Tip X	21.424	6.563	0.572	26.177	6.485	26.036	mm
Tip Y	-26.596	3.206	-32.204	-19.361	-31.874	-22.096	mm
Tip Z	178.996	1.049	176.567	180.579	177.319	180.444	mm
Tip jump Δ	7.711	32.769	0.219	180.949	0.317	5.153	mm

Total frames: 1227 | 3-D valid: 29 (2.4%) | Skeleton found: 1027 (83.7%) | Duration: 73.8 s | Mean throughput: 19.4 fps

Note: 'Tip jump' is frame-to-frame Euclidean distance of the smoothed tip in 3-D (mm). Only computed on consecutive valid frames.